

Module 1 - Frequency analysis

International Relations and Transboundary Cooperation

As an engineer you are called from the Swiss federal government to perform a preliminary evaluation on the impact of the Ill River regulation on the Rhein hydrological and suspended sediment regime.

The Ill River is the last major tributary of the Rhein before it enters Lake Constance. The Ill River catchment is entirely in Austria and its discharge at the confluence with the Rhein is partially regulated by the Illspitz Kraftwerk.

You are given discharge and suspended sediment concentration time series from the *Gisingen* ([eHYD](#)) measuring station on the Ill River and of the *Diepoldsau, Rietbrücke* ([BAFU](#)) station for the Rhein River, downstream of the confluence.



Data:

Gisingen

- 2003-2021: C suspended sediment concentration (g/L – 15min)
- 1976-2013: Q discharge (m³/s – 15min)

Diepoldsau, Rietbrücke

- 2012-2025: C suspended sediment concentration (g/L – 10 min)
- 1984-2025: Q discharge (m³/s – 10 min)

Analysis:

1. Data review: [3]

Create a timeseries with the maximum annual values of Q

Create a timeseries with the maximum monthly values of C

Then perform preliminary analysis: plot the boxplots of Q and C and compare the two stations (Q_{Gisingen} with Q_{Dipoldsau} and C_{Gisingen} with C_{Dipoldsau}). Discuss the

quantile values: do they have the same median? What can you say about the dataset by comparing the mean and the median? What is the IQR? How close do the distributions of each variable appear to be at the two sites?

2. Hypothesis testing: [5]

Test the hypothesis that each variable at the two sites comes from the same population: CGisingen vs CDipoldsau and QGisingen vs QDipoldsau. Choose the appropriate test and justify the choice. Use a 5% significance level.

3. Extreme value analysis: [5]

Analyze the extremes values of C and Q. For both stations fit a GEV to Q and C using the method of moments.

- a. Is the Gumbel distribution (EVI I) a good approximation for the GEV that you just fit? Provide a quantitative answer using the PPCC test
- b. Compare the GEV distributions between stations: QGisingen vs QDipoldsau and CGisingen vs CDipoldsau.

4. Return Period estimation: [5]

Estimate the Q and C values for a 10, 30, 50, 100, 300-year return period.

Estimate their standard error and the 90 % confidence interval. Analyse the reliability of your results for different return periods. To what extent does the record length affect the robustness of your extreme value estimates?

5. Comment: [2]

If you had to perform the analysis on the entire data series with the original time resolution, would the assumption of frequency analysis be valid?

Method:

The final product of this exercise will be an automatic code as a Jupyter notebook in *.ipynb* format to perform the analysis. Each of the steps (1 to 5) must be in a different section of the Jupyter notebook. Each section must be organized in the following way:

- Markdown cell: Title
- Markdown cell: List of variables
- Cell(s) where you write the functions that you will use for the analysis
- Cell (MAIN) where you run the analysis
- Cell (PLOT) following the main cell, where you plot the results
- Cell (PRINT) to print the results of the analysis, in nice format
- Markdown cell to comment on the results

At the start of the notebook import all the necessary packages and libraries for the analysis. The code for Module 1 exercise must be running from start to finish. I suggest making each section independent from the others, not compromising the entire output. A non-running code will not be graded.

Module 1 is a total of 20 points, each section with the score assigned in brackets []